OPTIMIZATION

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Lesson 5

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Remark

Second-order sufficient conditions work well in **low-dimensional spaces**. For higher-dimensional spaces, even finding points $x \in \mathbb{R}^n$ such that $\nabla f(x) = 0$ can be **extremely challenging**.

Key idea: introduce **numerical procedures** that allow us to compute solutions iteratively. These procedures are known as **optimization algorithms**.

- Most optimization algorithms generate a sequence of points such that, at each iteration k,

$$x^{k+1} = x^k + \alpha_k d^k,$$

  where $d^k \in \mathbb{R}^n$, with $||d^k|| = 1$ is the **search direction** (*where to move*), and $\alpha_k \in \mathbb{R}_+$ is the **step size** (*how far to move*).

- The **process is repeated** until:
  - the **gradient** is **close to zero**;
  - **no** significant **improvement** is observed.

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Why We Need Gradient Descent

Example

For example, consider the following function that needs to be minimized:

$$f(x) = x^2 \cdot \ln(x) - x \tag{1}$$

Setting the first derivative of this function to 0 yields the following condition:

$$f'(x) = 2x \cdot \ln(x) + x - 1 = 0.$$

This equation is hard to solve. By trial and error, one might get lucky and find out that $x = 1$ is indeed a solution to the first-order optimality condition because it satisfies $f'(1) = 2\ln(1) + 1 - 1 = 0$.

Furthermore, the second derivative $f''(x)$ can be shown to be positive at $x = 1$, so this point is at least a local minimum. However, solving an equation like this numerically causes all types of numerical and computational challenges; these types of challenges increase for multivariate optimization.

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Algorithms for Unconstrained Non Linear Programming

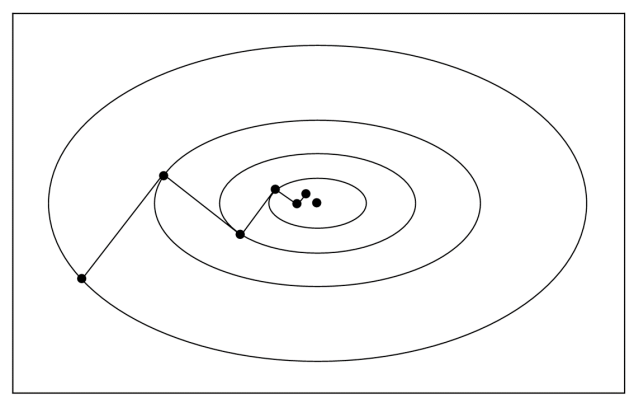
There are **two main approaches** to design optimization algorithms.

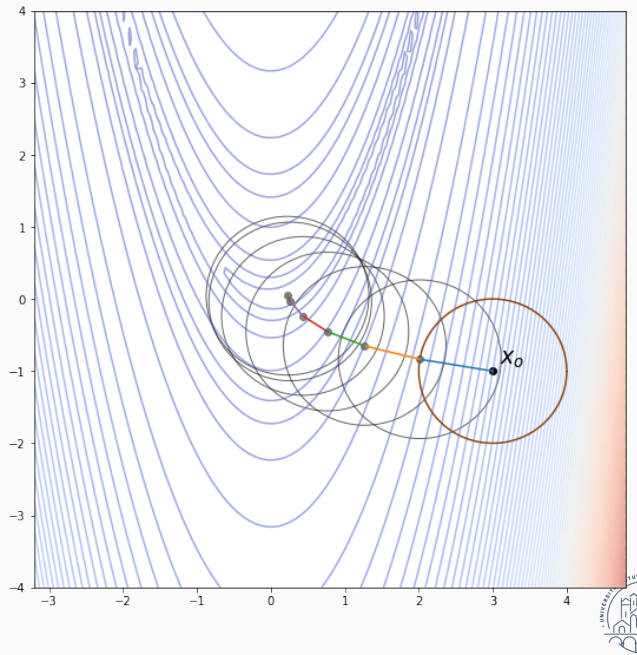
(1) **Line Search Methods:**

- determine a **direction** d^k;
- then compute a **step size** α_k;
- the new point lies along a line.

(2) **Trust Region Methods:**

- define a **region** around the current point;
- then compute the **best step** inside this region.





Key difference:

- **Line Search Methods:** from direction to step
- **Trust Region Methods:** from region (step) to direction

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We first introduce the definition of **descent direction**, which is fundamental for **Line Search Algorithms**.

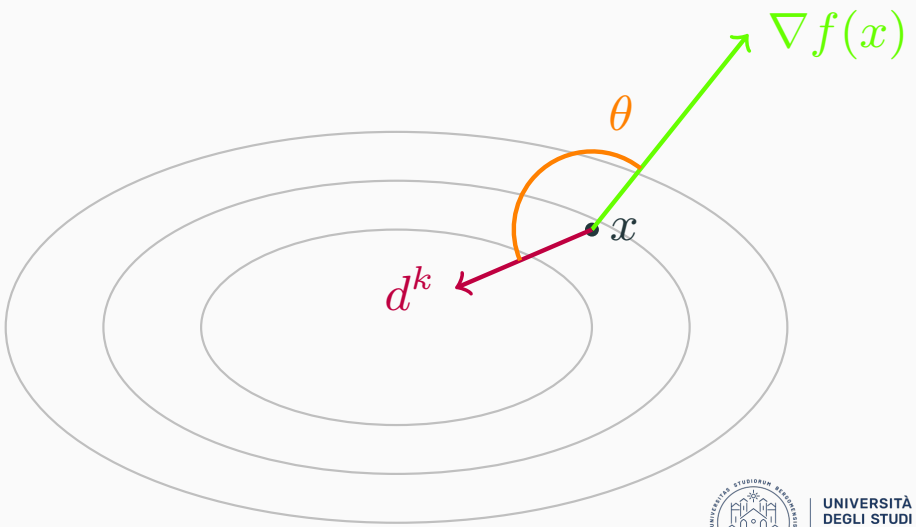
Definition (Descent Direction)

A direction $d \in \mathbb{R}^n$ is called **descent direction** if

$$\nabla f(x^k)^\top d^k < 0.$$

Geometric interpretation:

- recall that inner product $\nabla f(x^k)^\top d^k$ is equal to $\|\nabla f(x^k)\| \|d^k\| \cos\theta$, where θ is the angle between $\nabla f(x^k)$ and d^k;
- a **descent** direction forms an **obtuse angle** θ with the gradient;
- moving along d^k **locally decreases** the objective function.



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Line Search Algorithms: General Scheme

- At each iteration k, a **Line Search method** performs the following steps:

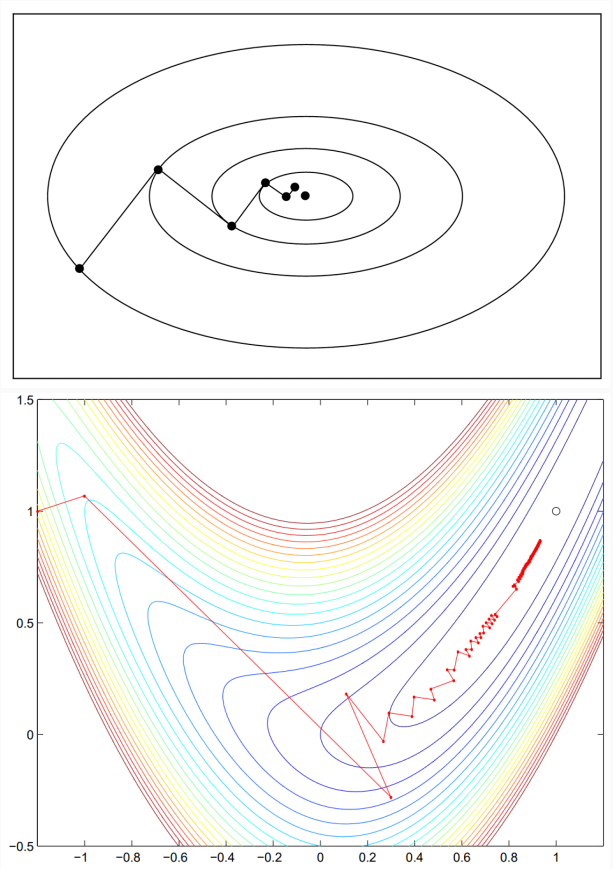
  (1) **check** if the current solution x^k is optimal by testing if $\nabla f(x^k) = 0$;

  (2) determine a **descent direction** d^k;

  (3) compute a **step size** α_k along direction d^k;

  (4) **update** the solution $x^{k+1} = x^k + \alpha_k d^k$ and iterate.

- **Stopping criterion:**
  - a **stationary point** is reached **or**
  - further **improvement** is **negligible**.



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Gradient Method:

- The **gradient method** is the simplest optimization algorithm and is based on the **first-order approximation** of the objective function f:

$$f(x^k + \alpha_k d^k) \approx f(x^k) + \alpha_k \nabla f(x^k)^T d^k,$$

  where $\alpha_k > 0$ and $\|d^k\| = 1$.

- Considering this expression as a function of d^k, to **get a minimum** we need to minimize $\nabla f(x^k)^T d^k$, neglecting $f(x^k)$ and α_k, as they are constant.
- **Minimizing** $\nabla f(x^k)^T d^k$ is equal to minimize $\|\nabla f(x^k)\| \cos\theta$, according to the definition of inner product, and given that $\|d^k\| = 1$.
- Therefore, the **minimum** is reached when $\cos\theta = -1$, which means that d^k should be **opposite** to the gradient and such that $\|d^k\| = 1$. Then, we get

$$d^k = \frac{-\nabla f(x^k)}{||\nabla f(x^k)||} \qquad \alpha_k = ||\nabla f(x^k)||.$$

- **Interpretation:** the algorithm moves in the direction of **steepest decrease**.

Gradient Method Example

Consider the function $f(x_1, x_2) = \frac{1}{2}x_1^2 + x_2^2$ with initial point $x^0 = (3, 2)^T$. Determine the next solution using the gradient algorithm with step size $\alpha = 0.5$.

- The gradient at x^0 is:

$$\nabla f(x^0) = \begin{pmatrix} 3 \\ 4 \end{pmatrix}.$$

- The steepest descent direction is:

$$d^0 = \frac{-\nabla f(x^0)}{\|\nabla f(x^0)\|} = -\frac{1}{5}\begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} -\frac{3}{5} \\ -\frac{4}{5} \end{pmatrix}.$$

- Assuming $\alpha = 0.5$, the next iterate is:

$$x^1 = x^0 + \alpha d^0 = \begin{pmatrix} 3 \\ 2 \end{pmatrix} + 0.5\begin{pmatrix} -\frac{3}{5} \\ -\frac{4}{5} \end{pmatrix} = \begin{pmatrix} \frac{27}{10} \\ \frac{8}{5} \end{pmatrix}.$$

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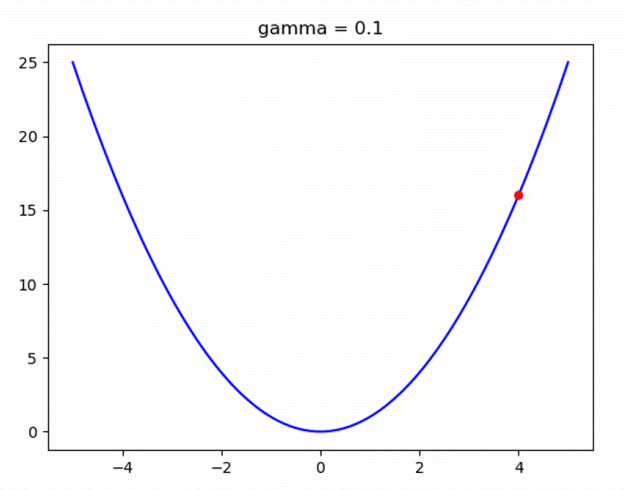
Why We Need Gradient Descent

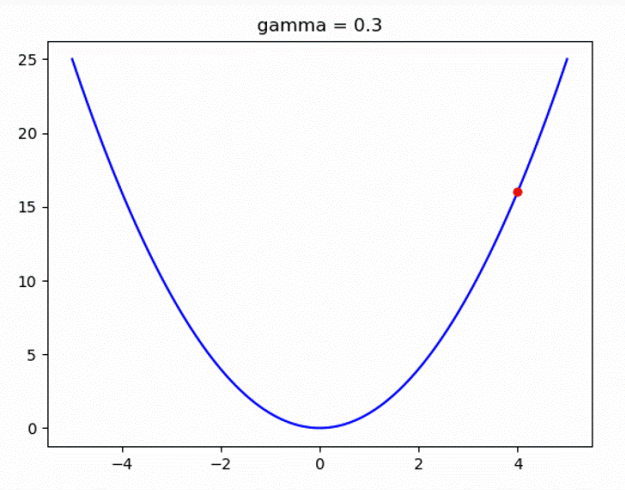
Gradient Method (Cauchy, 1847)

Let $x_0 \in \mathbb{R}$. For $t = 0, 1, \ldots$

$$x_{t+1} := x_t - \gamma f'(x_t),$$

with $\gamma \geq 0$ **learning rate**.

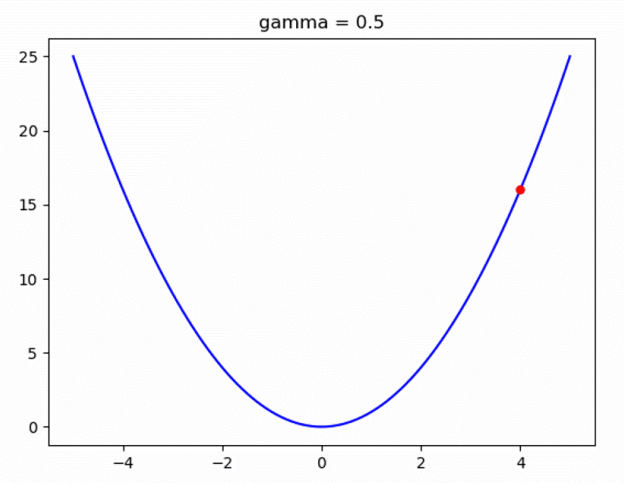


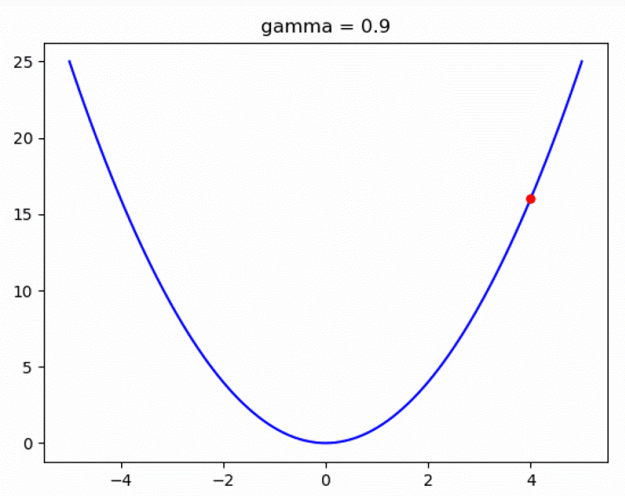


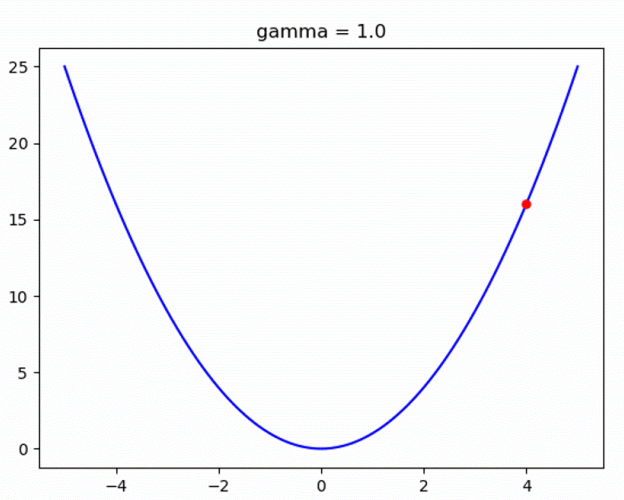
[0]Developed trying to solve the problem of determining the orbit of a celestial body from its equations of motion.

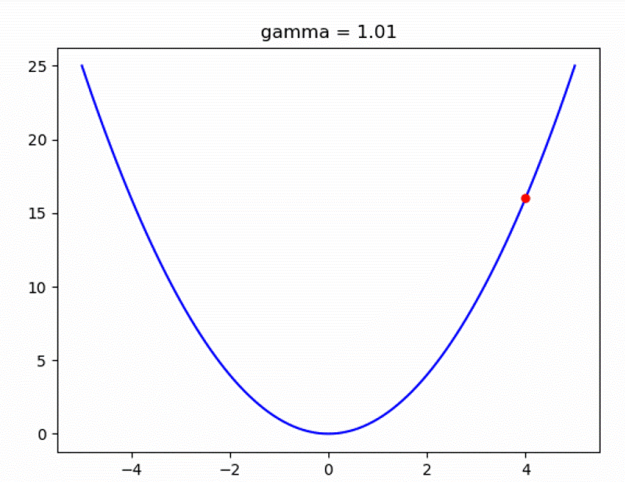
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Why We Need Gradient Descent









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Fresh Printing Results

ALGORITHMS

Risky Giant Steps Can Solve Optimization Problems Faster

8 |

New results break with decades of conventional wisdom for the gradient descent algorithm.



[1] https://www.quantamagazine.org/risky-giant-steps-can-solve-optimization-problems-faster-20230811/
Benjamin Grimmer (2023) Provably Faster Gradient Descent via Long Steps, arXiv.

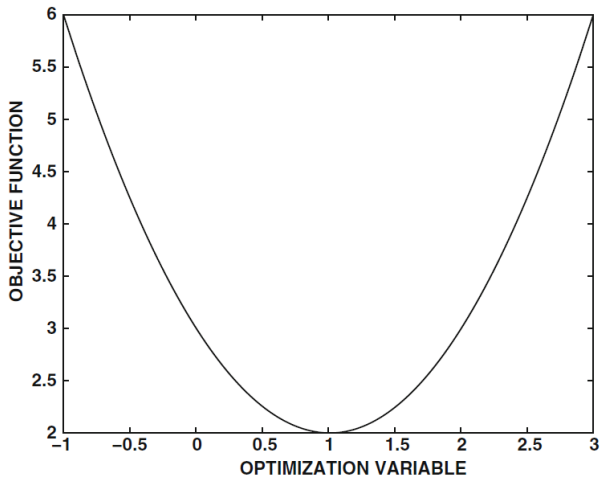
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The Divergence Problem

- Choosing a **very large learning rate** α can cause overshooting from the optimal solution, and even **divergence**. Consider the following example.

Example

Consider the quadratic function $f(x)$ in the following figure, which takes on its optimal value at $x = 1$:



(a) Single global minimum
$f(x) = x^2 - 2x + 3$

The Divergence Problem

Example

- Now imagine a situation where the starting point is $x_0 = 2$, and one chooses a large learning rate $\alpha = 10$.
- The derivative of $f(x) = 2x - 2$ evaluates to $f'(x_0) = f'(2) = 2$.
- Then, the update from the first step yields the following:

$$x_1 \Leftarrow x_0 - 10 \cdot 2 = 2 - 20 = -18.$$

- Note that the new point x_1 is much further away from the optimal value of $x = 1$, which is caused by the overshooting problem.
- Even worse, the absolute gradient is very large at this point, and it evaluates to $f'(-18) = -38$. If we keep the learning rate fixed, it will cause the solution to move at an even faster rate in the opposite direction:

$$x_2 \Leftarrow x_1 - 10 \cdot (-38) = -18 + 380 = 362.$$

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The Divergence Problem

Example

- In this case, the solution has overshot back in the original direction but is even further away from the optimal solution.
- Further updates cause back-and-forth movements at increasingly large amplitudes:

$$x_3 \Leftarrow x_2 - 10 \cdot 722 = 362 - 7220 = -6858, \quad x_4 \Leftarrow x_3 + 10 \cdot 13718 = 130322.$$

- Note that each iteration flips the sign of the current solution and increases its magnitude by a factor of about 20.
- In other words, the solution moves away faster and faster from an optimal solution until it leads to a numerical overflow.

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Gradient Descent: The Algorithm

Get near to a minimum $\mathbf{x}^*$ / close to the optimal value $f(\mathbf{x}^*)$?
Assumption: $f : \mathbb{R}^d \to \mathbb{R}$ convex, differentiable, has a global minimum $\mathbf{x}^*$.

Goal

Find $\mathbf{x}^* \in \mathbb{R}^d$ such that

$$f(\mathbf{x}) - f(\mathbf{x}^*) \leq \varepsilon.$$

Note that there can be several global minima $\mathbf{x}_1^* \neq \mathbf{x}_2^*$ with $f(\mathbf{x}_1^*) = f(\mathbf{x}_2^*)$.

Iterative Algorithm

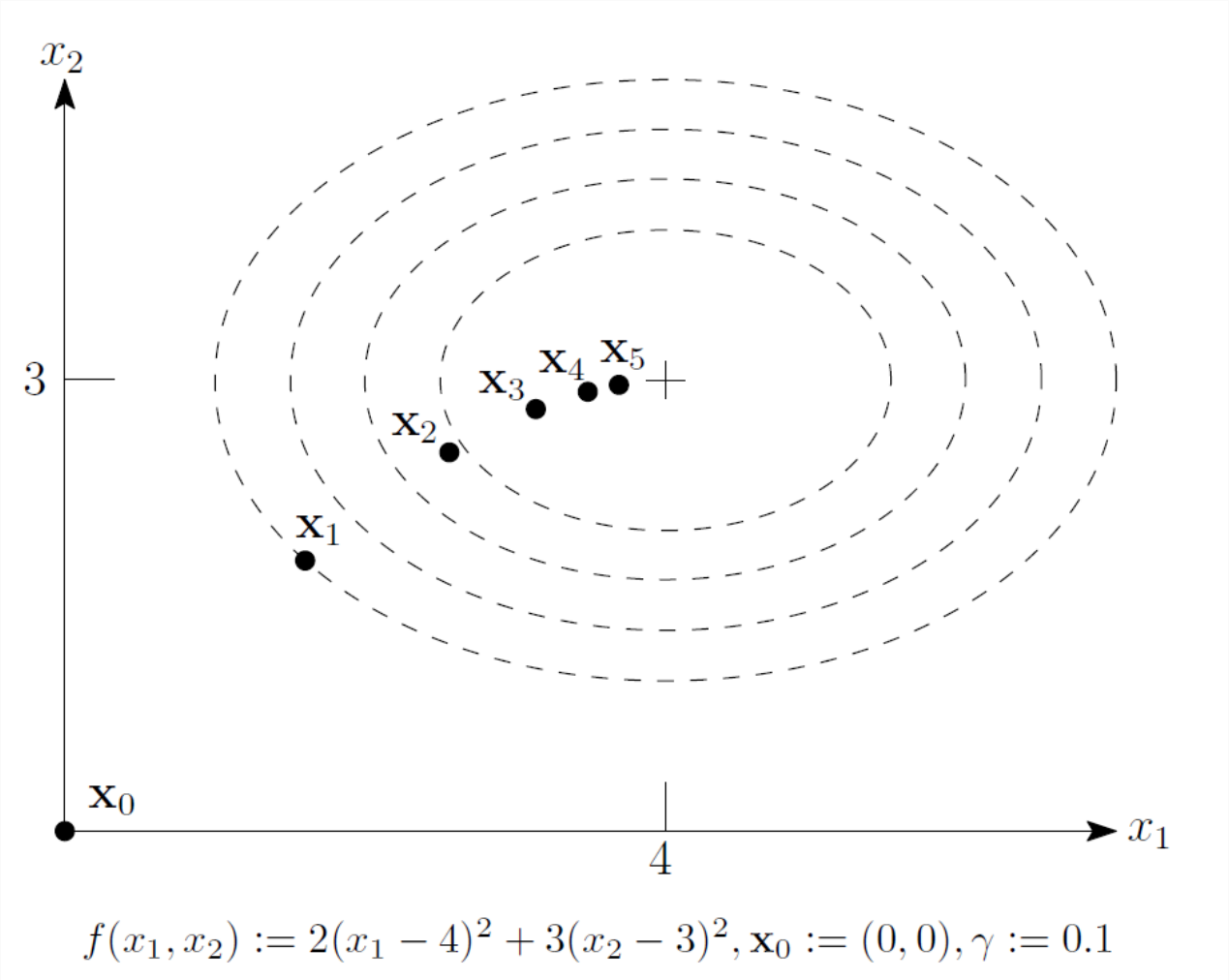
Choose $\mathbf{x}_0 \in \mathbb{R}^d$.

$$\mathbf{x}_{t+1} := \mathbf{x}_t - \gamma \nabla f(\mathbf{x}_t),$$

for timesteps $t = 0, 1, \ldots,$ and stepsize $\gamma \geq 0$.

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Example



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How to bound $f(\mathbf{x}_t) - f(\mathbf{x}^*)$?

- Abbreviate $\mathbf{g}_t := \nabla f(\mathbf{x}_t)$ (gradient descent: $\mathbf{g}_t = (\mathbf{x}_t - \mathbf{x}_{t+1})/\gamma$).

$$\mathbf{g}_t^\top (\mathbf{x}_t - \mathbf{x}^*) = \frac{1}{\gamma}(\mathbf{x}_t - \mathbf{x}_{t+1})^\top (\mathbf{x}_t - \mathbf{x}^*)$$

  .

- Apply $2\mathbf{v}^\top \mathbf{w} = ||\mathbf{v}||^2 + ||\mathbf{w}||^2 - ||\mathbf{v} - \mathbf{w}||^2$ to re-write:

$$\begin{aligned}\mathbf{g}_t^\top (\mathbf{x}_t - \mathbf{x}^*) =& \frac{1}{2\gamma}\left(||\mathbf{x}_t - \mathbf{x}_{t+1}||^2 + ||\mathbf{x}_t - \mathbf{x}^*||^2 - ||\mathbf{x}_{t+1} - \mathbf{x}^*||^2\right) \\ =& \frac{\gamma}{2}||\mathbf{g}_t||^2 + \frac{1}{2\gamma}\left(||\mathbf{x}_t - \mathbf{x}^*||^2 - ||\mathbf{x}_{t+1} - \mathbf{x}^*||^2\right).\end{aligned}$$

- Sum this up over the first T iterations:

$$\sum_{t=0}^{T-1} \mathbf{g}_t^\top (\mathbf{x}_t - \mathbf{x}^*) = \frac{\gamma}{2}\sum_{t=0}^{T-1} ||\mathbf{g}_t||^2 + \frac{1}{2\gamma}\left(||\mathbf{x}_0 - \mathbf{x}^*||^2 - ||\mathbf{x}_T - \mathbf{x}^*||^2\right).$$

Use first-order characterization of convexity: $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x})$ for all $\mathbf{x}, \mathbf{y}$.

- with $\mathbf{x} = \mathbf{x}_t$, $\mathbf{y} = \mathbf{x}^*$:

$$f(\mathbf{x}_t) - f(\mathbf{x}^*) \leq \mathbf{g}_t^\top (\mathbf{x}_t - \mathbf{x}^*)$$

  giving

$$\sum_{t=0}^{T-1} \Big(f(\mathbf{x}_t) - f(\mathbf{x}^*) \Big) \leq \frac{\gamma}{2} \sum_{t=0}^{T-1} ||\mathbf{g}_t||^2 + \frac{1}{2\gamma} ||\mathbf{x}_0 - \mathbf{x}^*||^2.$$

  an upper bound for the **average error** $f(\mathbf{x}_t) - f(\mathbf{x}^*)$ over the steps.

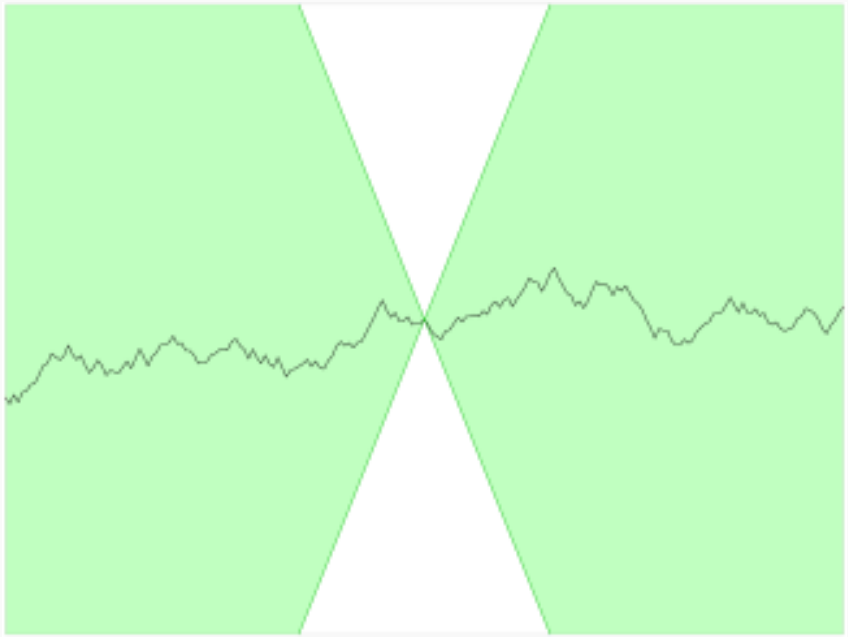
- Last iterate is not necessarily the best one.
- Stepsize is crucial.

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Lipschitz Convex Functions: $\mathcal{O}(1/\varepsilon^2)$ Steps

Assume that all gradients of f are bounded in norm.

- Equivalent to f being Lipschitz.
- Rules out many interesting functions (for example, the *supermodel* $f(x) = x^2$).



The Lipschitz condition (or Lipschitz continuity) ensures that your function always remains entirely outside the white cone, so it cannot e.g. become infinitely steep at one point.

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Lipschitz Convex Functions: $\mathcal{O}(1/\varepsilon^2)$ Steps

Theorem

Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex and differentiable with a global minimum $\mathbf{x}^*$; furthermore, suppose that $||\mathbf{x}_0 - \mathbf{x}^*|| \leq R$ and $||\nabla f(\mathbf{x})|| \leq B$ for all $\mathbf{x}$. Choosing the stepsize:

$$\gamma := \frac{R}{B\sqrt{T}}$$

gradient descent yields

$$\frac{1}{T}\sum_{t=0}^{T-1} f(\mathbf{x}_t) - f(\mathbf{x}^*) \leq \frac{RB}{\sqrt{T}}.$$

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Lipschitz Convex Functions: $\mathcal{O}(1/\varepsilon^2)$ Steps

Proof

- Plug $||\mathbf{x}_0 - \mathbf{x}^*|| \leq R$ and $||\nabla f(\mathbf{x})|| \leq B$ into Vanilla Analysis:

$$\sum_{t=0}^{T-1} \Big(f(\mathbf{x}_t) - f(\mathbf{x}^*) \Big) \leq \frac{\gamma}{2} \sum_{t=0}^{T-1} ||\mathbf{g}_t||^2 + \frac{1}{2\gamma} ||\mathbf{x}_0 - \mathbf{x}^*||^2 \leq \frac{\gamma}{2} B^2 T + \frac{1}{2\gamma} R^2.$$

- Choose γ such that

$$q(\gamma) = \frac{\gamma}{2} B^2 T + \frac{R^2}{2\gamma}$$

  is minimized.
- Solving $q'(\gamma) = 0$ yields the minimum $\gamma = \frac{R}{B\sqrt{T}}$ and $q\left(\frac{R}{B\sqrt{T}}\right) = RB\sqrt{T}$.
- Dividing by T, the result follows.

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Lipschitz Convex Functions: $\mathcal{O}(1/\varepsilon^2)$ Steps

$$T \geq \frac{R^2 B^2}{\varepsilon^2} \Rightarrow \text{ average error} \leq \frac{RB}{\sqrt{T}} \leq \varepsilon.$$

Advantages:

- dimension-independent (no d in the bound)!
- holds for both average, or best iterate.

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Smooth Functions

Definition

Let $f : dom(f) \to \mathbb{R}$ be differentiable, $X \subseteq dom(f)$, $L \in \mathbb{R}_+$. Function f is called **smooth** *(with parameter L) over X if*

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{L}{2}||\mathbf{x} - \mathbf{y}||^2, \quad \forall\, \mathbf{x}, \mathbf{y} \in X.$$

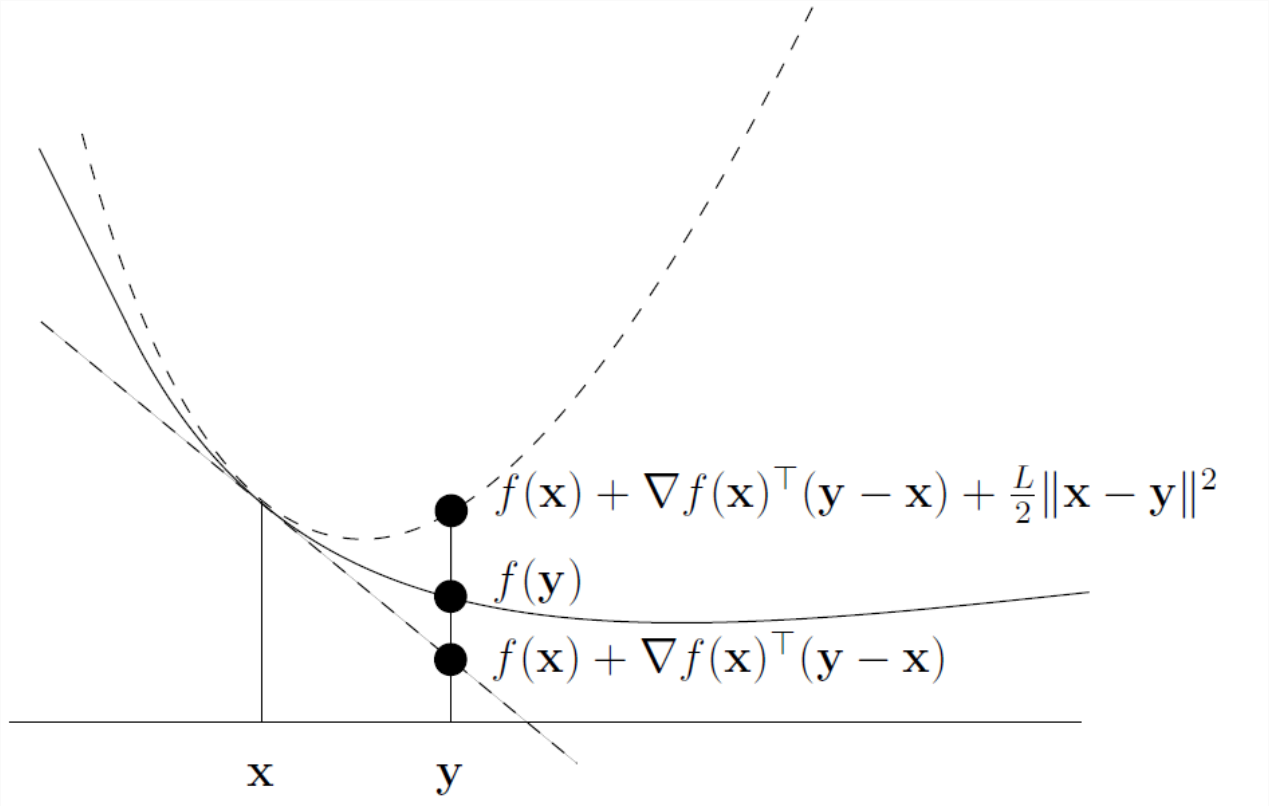
f smooth $:\iff$ smooth over $\mathbb{R}^d$.

Definition does not require convexity.

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Smooth Functions

Smoothness: For any $\mathbf{x}$, the graph of f is below a not too steep tangent paraboloid at $(\mathbf{x}, f(\mathbf{x}))$:



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Smooth Functions

- In general: quadratic functions are smooth.
- Operations that preserve smoothness (the same that preserve convexity).

Lemma

Let $f_1, \ldots, f_m$ be functions that are smooth with parameters $L_1, \ldots, L_m$ and let $\lambda_1, \ldots, \lambda_m \in \mathbb{R}_+$. Then, the function $f := \sum_{i=1}^m \lambda_i f_i$ is smooth with parameter $\sum_{i=1}^m \lambda_i L_i$.

Lemma

Let f be smooth with parameter L, and let $g(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$, for $A \in \mathbb{R}^{d \times m}$ and $\mathbf{b} \in \mathbb{R}^d$. Then the function $f \circ g$ is smooth with parameter $L||A||^2$, where $||A||$ is the spectral norm of A.

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Smooth vs Lipschitz

- Bounded gradients $\iff$ Lipschitz continuity of f.
- Smoothness $\iff$ Lipschitz continuity of ∇f (in the convex case).

Lemma

Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex and differentiable. The following two statements are equivalent.

1. f is smooth with parameter L.
2. $||\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})|| \leq L||\mathbf{x} - \mathbf{y}||$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$.

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Sufficient Decrease

Lemma

Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and smooth with parameter L. With stepsize

$$\gamma := \frac{1}{L},$$

gradient descent satisfies

$$f(\mathbf{x}_{t+1}) \leq f(\mathbf{x}_t) - \frac{1}{2L}||\nabla f(\mathbf{x}_t)||^2, \quad t \geq 0.$$

Remark

More specifically, this already holds if f is smooth with parameter L over the line segment connecting $\mathbf{x}_t$ and $\mathbf{x}_{t+1}$.

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Sufficient Decrease

Proof

Use smoothness and definition of gradient descents ($\mathbf{x}_{t+1} - \mathbf{x}_t = -\nabla f(\mathbf{x}_t)/L$):

$$\begin{aligned} f(\mathbf{x}_{t+1}) &\leq f(\mathbf{x}_t) + \nabla f(\mathbf{x}_t)^\top (\mathbf{x}_{t+1} - \mathbf{x}_t) + \frac{L}{2}||\mathbf{x}_t - \mathbf{x}_{t+1}||^2 \\ &= f(\mathbf{x}_t) - \frac{1}{L}||\nabla f(\mathbf{x}_t)||^2 + \frac{1}{2L}||\nabla f(\mathbf{x}_t)||^2 \\ &= f(\mathbf{x}_t) - \frac{1}{2L}||\nabla f(\mathbf{x}_t)||^2. \end{aligned}$$

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Smooth Convex Functions: $\mathcal{O}(1/\varepsilon^2)$ Steps

Theorem

Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex and differentiable with a global minimum $\mathbf{x}^*$. Furthermore, suppose that f is smooth with parameter L. Choosing stepsize

$$\gamma := \frac{1}{L}$$

gradient descent yields

$$f(\mathbf{x}_T) - f(\mathbf{x}^*) \leq \frac{L}{2T}||\mathbf{x}_0 - \mathbf{x}^*||^2, \quad T > 0.$$

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Smooth Convex Functions: $\mathcal{O}(1/\varepsilon^2)$ Steps

Proof

Vanilla Analysis:

$$\sum_{t=0}^{T-1} \Big(f(\mathbf{x}_t) - f(\mathbf{x}^*) \Big) \leq \frac{\gamma}{2} \sum_{t=0}^{T-1} ||\nabla f(\mathbf{x}_t)||^2 + \frac{1}{2\gamma} ||\mathbf{x}_0 - \mathbf{x}^*||^2.$$

This time, we can bound the squared gradients by sufficient decrease:

$$\frac{1}{2L} \sum_{t=0}^{T-1} ||\nabla f(\mathbf{x}_t)||^2 \leq \sum_{t=0}^{T-1} \Big(f(\mathbf{x}_t) - f(\mathbf{x}_{t+1}) \Big) = f(\mathbf{x}_0) - f(\mathbf{x}_T).$$

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Smooth Convex Functions: $\mathcal{O}(1/\varepsilon^2)$ Steps

Proof

Putting it together with $\gamma = 1/L$:

$$\sum_{t=0}^{T-1} \Big(f(\mathbf{x}_t) - f(\mathbf{x}^*) \Big) \leq \frac{1}{2L} \sum_{t=0}^{T-1} ||\nabla f(\mathbf{x}_t)||^2 + \frac{L}{2} ||\mathbf{x}_0 - \mathbf{x}^*||^2$$

$$\leq f(\mathbf{x}_0) - f(\mathbf{x}_T) + \frac{L}{2} ||\mathbf{x}_0 - \mathbf{x}^*||^2.$$

Rewriting:

$$\sum_{t=1}^{T} \Big(f(\mathbf{x}_t) - f(\mathbf{x}^*) \Big) \leq \frac{L}{2} ||\mathbf{x}_0 - \mathbf{x}^*||^2.$$

As last iterate is the best (sufficient decrease!):

$$f(\mathbf{x}_T) - f(\mathbf{x}^*) \leq \frac{1}{T} \sum_{t=1}^{T} \Big(f(\mathbf{x}_t) - f(\mathbf{x}^*) \Big) \leq \frac{L}{2T} ||\mathbf{x}_0 - \mathbf{x}^*||^2.$$

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Smooth Convex Functions: $\mathcal{O}(1/\varepsilon^2)$ Steps

$R^2 := ||\mathbf{x}_0 - \mathbf{x}^*||^2$.

$$T \geq \frac{R^2 L}{2\varepsilon} \Rightarrow \text{ error} \leq \frac{L}{2T} R^2 \leq \varepsilon.$$

- $50 \cdot R^2 L$ iterations for error 0.01 ...
- ... as opposed to $10{,}000 \cdot R^2 B^2$ in the Lipschitz case.

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Assignment

1. Compute the gradient of the following functions:
   - $F = \mathbf{a}^\top \mathbf{x}$;
   - $F = \mathbf{x}^\top A\mathbf{x}$ with A symmetric;
   - $F = \mathbf{x}^\top A\mathbf{x}$ for a generic matrix A;
   - $F = (\mathbf{a}^\top \mathbf{x})^2$;
   - $F = \frac{1}{2n}||A\mathbf{x} - \mathbf{b}||^2$.
2. Prove that the quadratic function $f(\mathbf{x}) = \mathbf{x}^\top Q\mathbf{x} + \mathbf{b}^\top \mathbf{x} + c$, with Q symmetric is smooth with parameter $2||Q||$.
3. Apply 2 iterations of the gradient descent method to the function $F(x_1, x_2) = (x_1 - 1)^2 + x_2^3 - x_1 x_2$ with $\mathbf{x}_0 = (1, 1)$.

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